ASSIGNMENT II — CASE STUDY

EXPLORATORY DATA ANALYSIS AND VISUALIZATION OF A REAL-WORLD DATASET

CREDIT CARD FRAUD DETECTION USING EXPLORATORY DATA ANALYSIS, DATA VISUALIZATION AND DEEP LEARNING

Particulars	Details
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1. ABSTRACT

Credit card fraud detection is an important application of data analytics and machine learning in the financial sector. This project investigates a real-world credit card transaction dataset using Exploratory Data Analysis (EDA), statistical analysis, data visualization, and deep learning. The dataset contains transaction information, anonymized numerical features, transaction amounts, and a binary fraud label. EDA is performed to examine data quality, class imbalance, feature distributions, outliers, and relationships among variables. The data is preprocessed using stratified splitting and feature standardization. A Multilayer Perceptron (MLP) neural network is trained using class weights to address the imbalance between legitimate and fraudulent transactions. Model performance is evaluated using precision, recall, F1-score, ROC-AUC, average precision, and a confusion matrix. The project demonstrates how statistical exploration and deep learning can support the analysis of fraudulent financial transactions.

2. INTRODUCTION AND OBJECTIVES

2.1 Introduction

The increasing use of credit cards and digital payment systems has made electronic transactions an essential part of modern financial activities. However, unauthorized transactions and fraudulent activities create financial losses for customers, merchants, and financial institutions.

Credit card fraud detection involves identifying suspicious transactions and distinguishing them from legitimate transactions. This is a challenging classification problem because fraudulent transactions typically represent only a small proportion of all transactions.

Traditional rule-based systems identify fraud using predefined conditions. However, such systems may struggle to capture complex patterns in transaction data. Data analytics and machine learning provide alternative approaches for identifying relationships and patterns associated with fraudulent behavior.

Exploratory Data Analysis helps understand the structure, quality, and distribution of transaction data before model development. Statistical summaries and visualizations reveal class imbalance, unusual observations, feature relationships, and possible patterns that may be useful for classification.

Deep learning techniques can learn complex relationships between input features and target labels. In this project, a Multilayer Perceptron neural network is developed to classify credit card transactions as legitimate or fraudulent.

The project combines data exploration, statistical analysis, visualization, preprocessing, deep learning, and performance evaluation into a single analytical workflow.

2.2 Problem Statement

Credit card transaction datasets contain a large number of legitimate transactions and a comparatively small number of fraudulent transactions. Detecting the minority fraud class accurately while minimizing false alerts is a major challenge.

The problem addressed in this project is to analyze transaction data, identify meaningful statistical patterns, and develop a deep learning classification model that estimates whether a transaction is fraudulent.

2.3 Objectives

The major objectives of this project are:

1. To select and analyze a real-world credit card transaction dataset containing more than 15 features.
2. To examine dataset dimensions, data types, missing values, duplicate records, and statistical summaries.
3. To perform univariate, bivariate, and multivariate exploratory analysis.
4. To create at least five meaningful visualizations and interpret the patterns observed.

5. To analyze continuous and discrete variables and investigate feature relationships.
6. To preprocess the data using stratified splitting and standardization.
7. To develop an MLP deep learning model for binary classification.
8. To perform basic hyperparameter tuning using validation data.
9. To evaluate the model using suitable classification metrics and result visualizations.
10. To document findings, limitations, and future improvements.

2.4 Scope of the Project

This project focuses on offline analysis of the publicly available Credit Card Fraud Detection dataset. It examines transaction patterns, develops a neural network classifier, and evaluates its performance on a held-out test set.

The project demonstrates the analytical workflow required for fraud detection. It does not constitute a production-ready financial fraud prevention system, since real-time integration, transaction authorization, and deployment are outside the current scope.

3. DATASET DESCRIPTION

3.1 Dataset Source

The dataset used in this project is the Credit Card Fraud Detection dataset published through Kaggle by the Machine Learning Group at Université Libre de Bruxelles (ULB).

Dataset link: https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud

The dataset contains transactions made by European cardholders during two days in September 2013.

3.2 Dataset Size

The dataset contains 284,807 transaction records and 31 columns, including 30 input features and one target variable.

Property	Description
Dataset name	Credit Card Fraud Detection
Data type	Structured numerical data
Total records	284,807
Total columns	31
Input features	30
Target variable	Class
Problem type	Binary classification
Legitimate class	0
Fraudulent class	1

3.3 Feature Description

The dataset contains the following categories of features.

Feature	Description
Time	Seconds elapsed between a transaction and the dataset's reference point
V1–V28	Anonymized numerical features produced using Principal Component Analysis (PCA)

Feature	Description
Amount	Monetary value of the transaction
Class	Target label indicating legitimate or fraudulent transactions

The `Time` and `Amount` features provide transaction-level information. The features V1 through V28 are transformed numerical variables. Their original business meanings are not directly available.

The `Class` variable is the target used to train and evaluate the binary classification model.

3.4 Target Variable

The target variable contains two classes:

- **Class 0:** Legitimate transaction.
- **Class 1:** Fraudulent transaction.

The distribution of these classes must be examined because a large imbalance can affect model training and evaluation.

3.5 Software and Libraries

The project is implemented in Python using Google Colab.

The primary libraries used are:

Library	Purpose
Pandas	Data loading, manipulation, and statistical summaries
NumPy	Numerical operations
Matplotlib	Visualization
Seaborn	Statistical plots and heatmaps
Scikit-learn	Data splitting, scaling, class weights, and evaluation
TensorFlow/Keras	Deep learning model development and training

4. EXPLORATORY DATA ANALYSIS AND PREPROCESSING

4.1 Exploratory Data Analysis

Exploratory Data Analysis is the process of examining data using statistical summaries and visualizations to understand its structure, distributions, relationships, and potential quality issues.

EDA was performed before training the model to understand the dataset and identify important characteristics of credit card transactions.

The analysis includes:

- Dataset dimensions and data types.
- Missing-value analysis.
- Duplicate-record analysis.
- Univariate statistical analysis.
- Distribution and outlier analysis.
- Class imbalance analysis.
- Bivariate and multivariate analysis.

- Correlation analysis.

4.2 Data Quality Analysis

4.2.1 Missing Values

Missing values were examined across all dataset columns using Pandas.

Missing-value analysis helps determine whether observations contain incomplete information that may affect statistical analysis or model training.

Observed result: [Insert the total missing-value count from Colab.]

If missing values are present, an appropriate strategy such as imputation or removal should be selected based on the nature and extent of the missing data. The selected treatment must be documented.

4.2.2 Duplicate Records

Duplicate records were examined to assess the quality of the dataset.

Duplicate analysis is important because repeated records can influence statistical summaries and model evaluation. However, repeated transactions are not automatically errors and should be reviewed before removal.

Observed result: [Insert the duplicate-row count from Colab.]

Any duplicate-removal decision should be documented, including the number of rows removed, if applicable.

4.2.3 Data Types

The data types of the columns were inspected using Pandas.

The dataset primarily contains numerical features, which makes it suitable for statistical analysis, numerical visualization, and neural network classification.

The target variable is binary and is used to distinguish legitimate transactions from fraudulent ones.

4.3 Univariate Statistical Analysis

Univariate analysis examines one variable at a time. It helps understand the central tendency, spread, frequency, and distribution of individual features.

Descriptive statistics were generated using the `describe()` function.

The statistical measures include:

- **Count:** Number of non-missing observations.
- **Mean:** Average value of a feature.
- **Standard deviation:** Measure of variability.
- **Minimum and maximum:** Lowest and highest observed values.
- **Quartiles:** Values representing the 25th, 50th, and 75th percentiles.

These measures help identify the scale of numerical variables and potential unusual observations.

The transaction amount distribution was studied using a histogram, while a boxplot was used to investigate the spread and potential outliers.

Observed findings: [Insert the actual mean, median, spread, and distribution observations obtained from your notebook.]

4.4 Discrete and Continuous Distribution Analysis

Distribution analysis helps understand the frequency and probability patterns of the variables in the dataset.

4.4.1 Discrete Distribution

The target variable `Class` is categorical and binary, with two possible values: 0 and 1.

Its frequency distribution was examined using a bar chart. The relative frequencies were calculated as percentages to compare the proportions of legitimate and fraudulent transactions.

For a binary variable, the class proportion can be expressed as:

$$P(Class = 1) = \frac{Number\ of\ fraud\ cases}{Total\ transactions}$$

Similarly,

$$P(Class = 0) = \frac{Number\ of\ legitimate\ cases}{Total\ transactions}$$

These proportions provide a quantitative description of class imbalance.

4.4.2 Continuous Distribution

The `Amount` feature is a continuous numerical variable representing transaction values.

Its distribution was analyzed using a histogram and boxplot. These plots help identify the concentration of values, spread, skewness, and possible extreme observations.

Observed findings: [Describe the actual shape of the histogram and the observed class proportions.]

4.5 Outlier Analysis

Outlier analysis was performed using the interquartile range method.

The interquartile range is calculated as:

$$IQR = Q_3 - Q_1$$

The conventional lower and upper bounds are:

$$L = Q_1 - 1.5(IQR)$$
$$U = Q_3 + 1.5(IQR)$$

Observations below the lower bound or above the upper bound are flagged as potential outliers.

A boxplot of `Amount` was used to visualize the spread and potential extreme values. A second boxplot compared transaction amounts across the two target classes.

Outliers were not automatically treated as errors because unusually large or small transactions may be legitimate. The analysis therefore focuses on identifying and understanding these observations.

Observed result: [Insert the actual IQR bounds and potential outlier count from Colab.]

4.6 Bivariate Analysis

Bivariate analysis investigates relationships between two variables.

In this project, transaction amount was compared with the target class to determine whether the distributions of legitimate and fraudulent transactions differed.

The `Time` and `Amount` features were also compared using a scatter plot. This visualization helps examine how transaction amounts vary across the recorded time period and how fraudulent transactions appear within the sampled observations.

The analysis of selected features V1, V3, V4, and V14 was used to compare their distributions between the two classes.

Observed findings: [Describe the actual differences and overlaps visible in these plots.]

4.7 Multivariate Analysis

Multivariate analysis examines relationships among multiple variables simultaneously.

A correlation matrix was generated for the numerical columns, followed by a heatmap visualization.

Pearson correlation coefficients range from -1 to +1:

- Values close to +1 indicate a strong positive linear association.
- Values close to -1 indicate a strong negative linear association.
- Values close to 0 indicate a weak linear association.

The heatmap helps identify potentially related features and examine their correlations with the target variable.

Correlation does not establish causation, and the anonymized features cannot automatically be assigned their original business interpretations.

Observed findings: [Identify the strongest actual correlations visible in the heatmap.]

4.8 Data Preprocessing

Preprocessing was performed to prepare the dataset for deep learning.

4.8.1 Feature and Target Separation

The `Class` column was separated as the target variable, while the remaining 30 columns were retained as model input features.

4.8.2 Training, Validation, and Test Split

The dataset was divided into three subsets using stratified sampling:

Subset	Approximate proportion	Purpose
Training	70%	Learn model parameters
Validation	15%	Monitor performance and compare hyperparameters
Test	15%	Final evaluation on held-out data

Stratification was used to preserve the class proportions across the subsets.

4.8.3 Feature Standardization

StandardScaler was used to standardize the numerical input features.

The transformation is:

$$z = \frac{x - \mu}{\sigma}$$

where x is the original value, μ is the training-set mean, and σ is the training-set standard deviation.

The scaler was fitted only on the training set. The same transformation was then applied to the validation and test sets to avoid data leakage.

4.8.4 Handling Class Imbalance

The dataset contains substantially different numbers of legitimate and fraudulent transactions.

Balanced class weights were calculated using the training labels. These weights increase the contribution of minority-class errors to the training loss without altering the validation or test distributions.

4.8.5 Preprocessing Summary

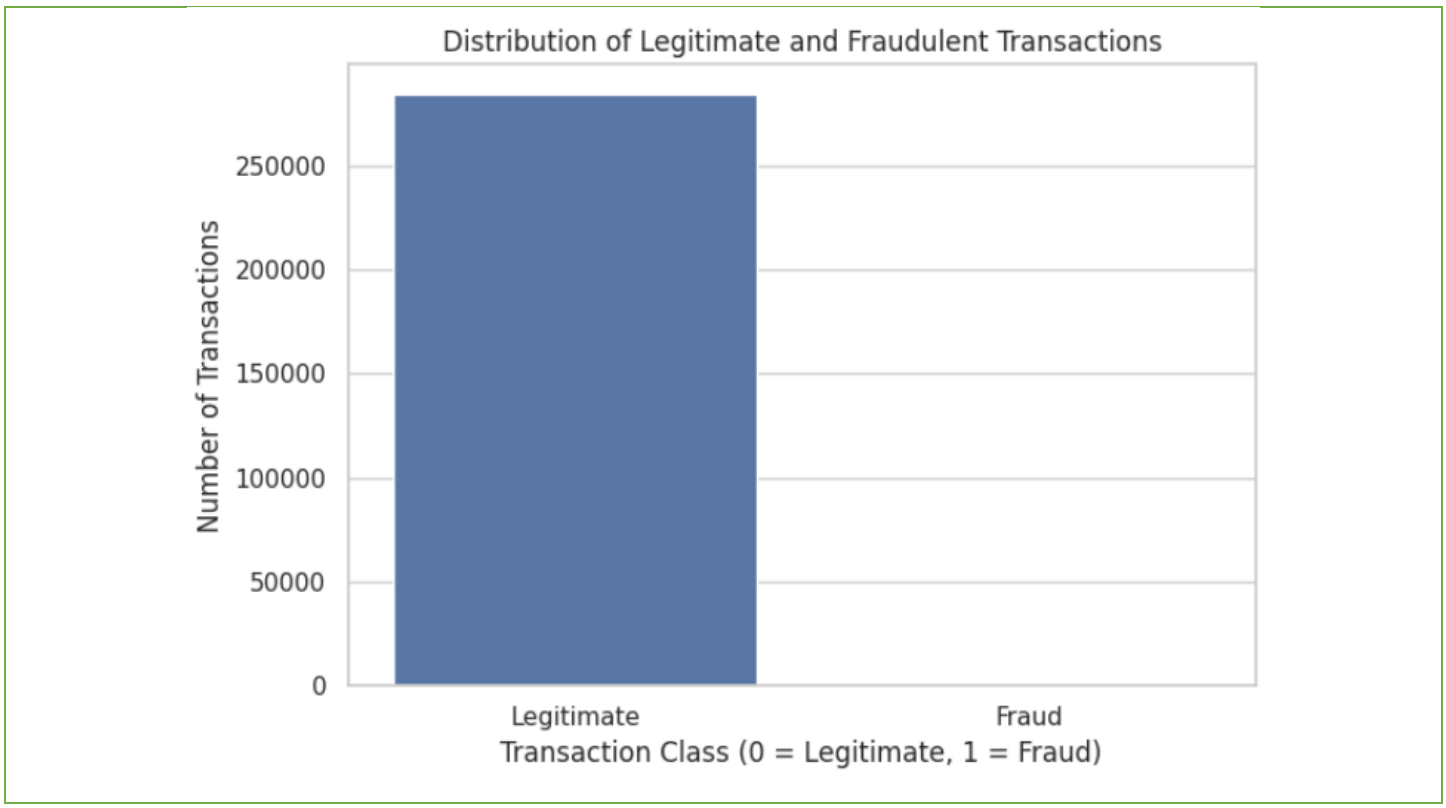
The preprocessing stage prepared the numerical features for neural network training, maintained separate validation and test sets, and incorporated a class-imbalance strategy.

5. DATA VISUALIZATION AND INTERPRETATION

At least five EDA visualizations were created to explore the transaction dataset. Each visualization serves a different analytical purpose.

5.1 Visualization 1: Distribution of Legitimate and Fraudulent Transactions

Figure 1: Class Distribution of Credit Card Transactions



Purpose: To compare the frequency of legitimate and fraudulent transactions.

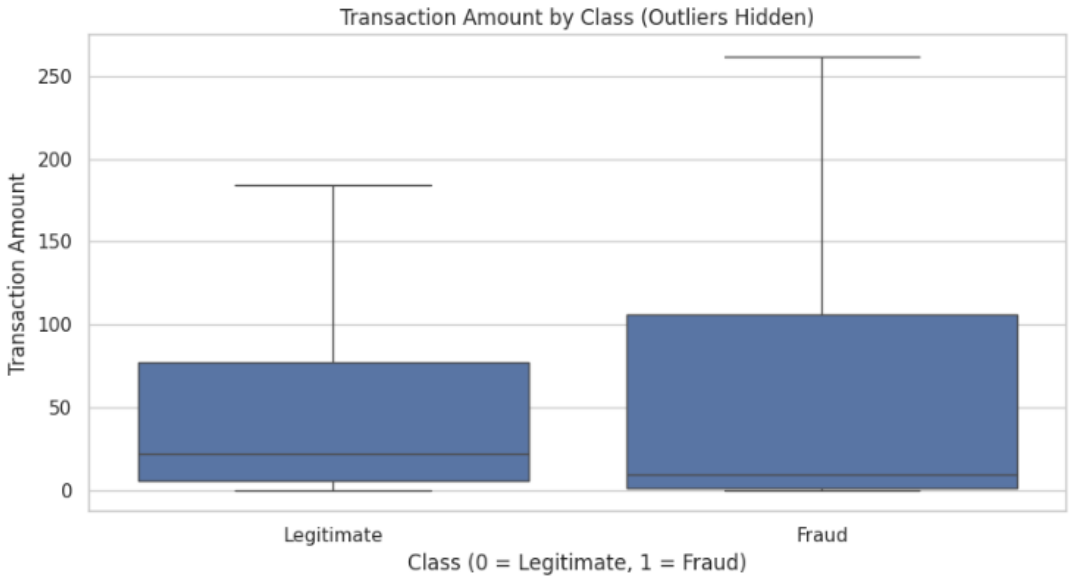
Method: A bar chart was generated using the `Class` column.

Interpretation: The chart compares the counts of Class 0 and Class 1. The observed proportions demonstrate the level of class imbalance in the dataset. This finding is important when selecting training strategies and evaluation metrics.

Observed finding: [Insert the actual class counts and percentages.]

5.2 Visualization 2: Distribution of Transaction Amounts

Figure 2: Histogram of Transaction Amounts



Class	count	mean	median	std	min	max
0	284315	88.29	22.00	250.11	0.0	25691.16
1	492	122.21	9.25	256.68	0.0	2125.87

Purpose: To understand the distribution of transaction values.

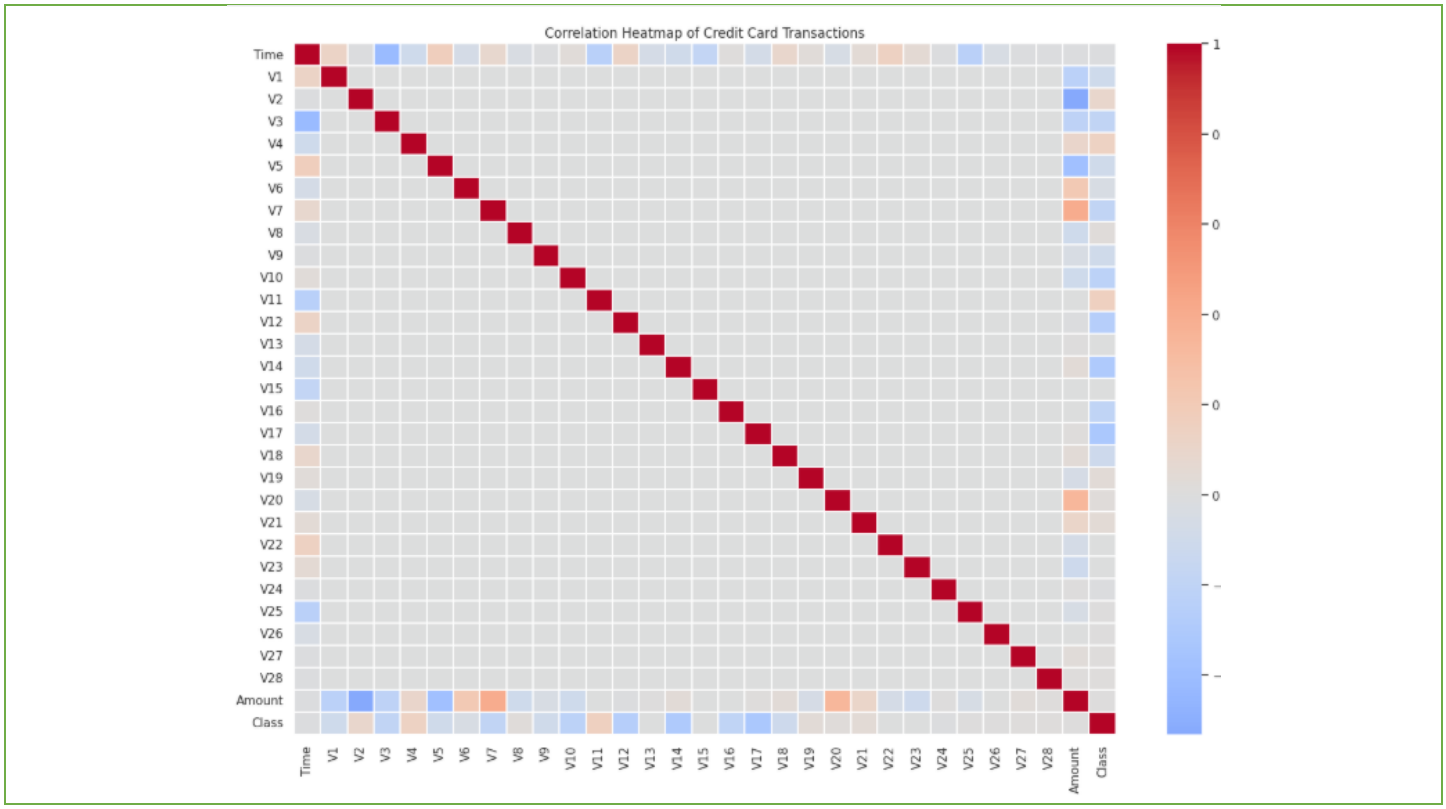
Method: A histogram with 50 bins and a kernel density estimate was created.

Interpretation: The histogram illustrates the concentration and spread of transaction amounts. Its shape can reveal skewness, clusters of observations, and the presence of unusually large values.

Observed finding: [Describe the actual distribution shown in your plot.]

5.3 Visualization 3: Correlation Heatmap

Figure 3: Correlation Heatmap of Numerical Features



Purpose: To identify linear associations among numerical features and the target variable.

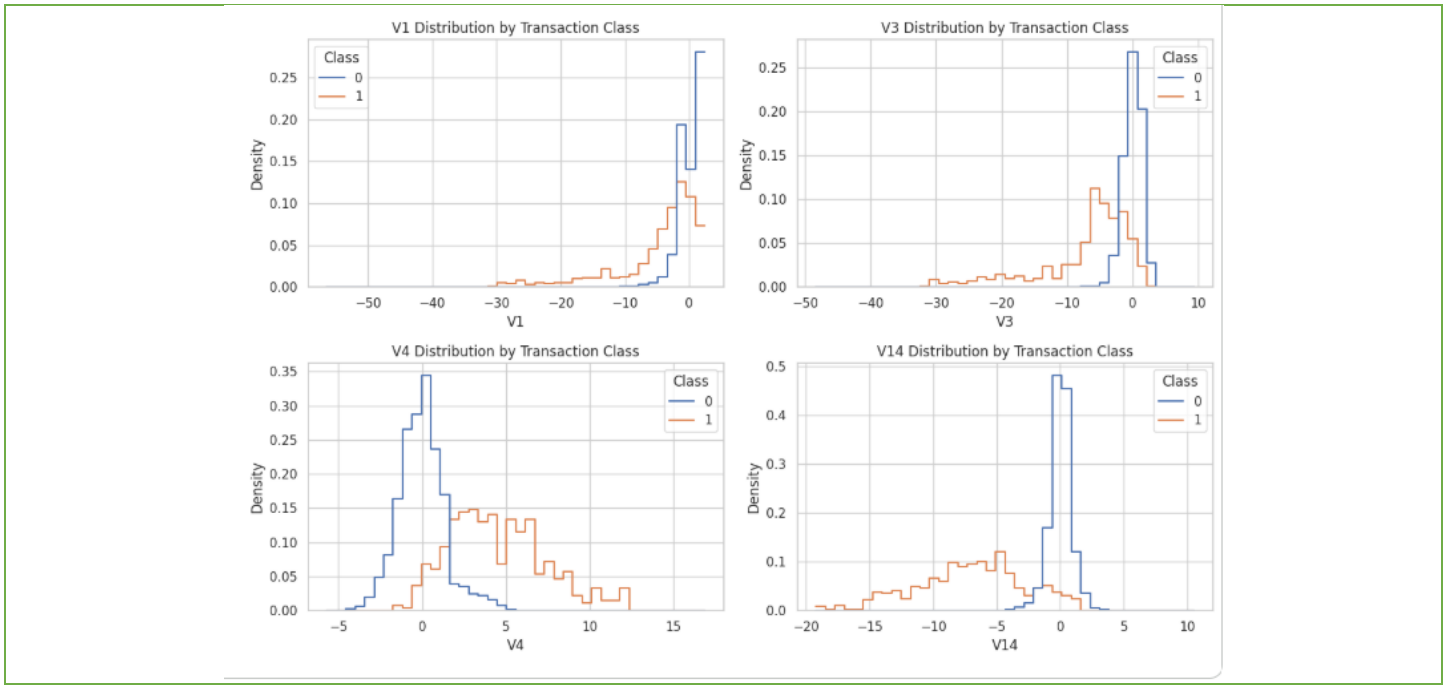
Method: Pearson correlations were calculated and visualized using a heatmap.

Interpretation: Strong positive or negative correlations indicate linear relationships between variables. Correlations with the target may help identify features associated with class differences, although they do not prove causation.

Observed finding: [Insert the actual strongest correlations and their directions.]

5.4 Visualization 4: Feature Distributions by Class

Figure 4: Distribution of Selected Features by Transaction Class



Purpose: To compare selected feature distributions for legitimate and fraudulent transactions.

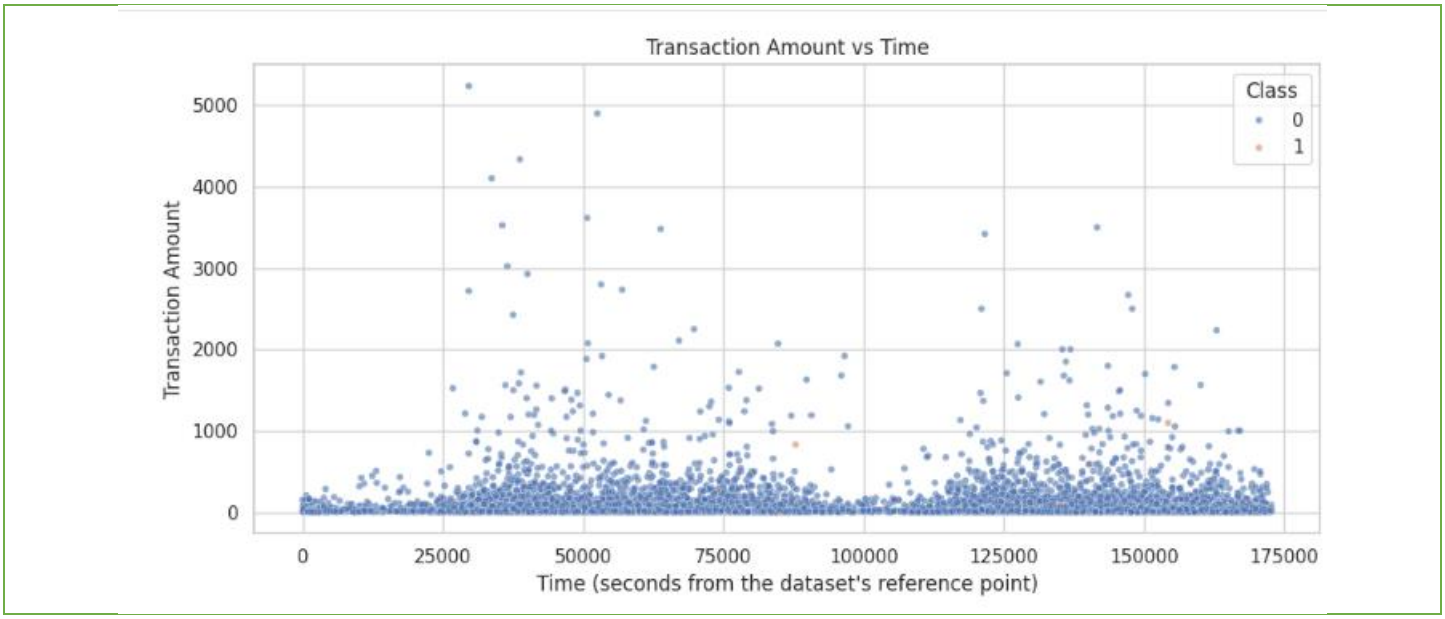
Method: Distribution plots were created separately for both classes.

Interpretation: Differences in the shapes, central regions, and overlap of the distributions indicate whether the features exhibit different statistical patterns across the two classes.

Observed finding: [Describe the differences you actually observe.]

5.5 Visualization 5: Transaction Amount versus Time

Figure 5: Transaction Amount versus Recorded Transaction Time



Purpose: To examine the relationship between transaction amount and recorded time.

Method: A reproducible random sample of up to 10,000 observations was plotted, with points colored by class.

Interpretation: The scatter plot displays the spread of transaction amounts over the recorded time period and the positions of fraudulent transactions within the sample. It helps identify visible clusters or unusual patterns without assuming that a visual association is statistically significant.

5.8 Summary of Visualization Insights

The visualizations collectively examine class frequency, transaction amount distributions, potential outliers, feature-level differences, time-related patterns, and multivariate correlations.

These analyses provide an exploratory understanding of the dataset and help motivate the use of class weighting and multiple evaluation metrics during model development.

The final conclusions must reflect the patterns actually observed in the generated plots.

6. DEEP LEARNING MODEL

6.1 Model Selection

A Multilayer Perceptron (MLP) was selected for binary classification.

An MLP is a feed-forward neural network consisting of an input layer, hidden layers, and an output layer. It learns numerical relationships through weighted connections and nonlinear activation functions.

The MLP is appropriate for this project because the dataset contains structured numerical features and a binary target variable.

6.2 Model Architecture

The implemented network contains the following layers:

Layer	Configuration	Purpose
Input	30 features	Receives transaction data
Dense layer 1	64 neurons, ReLU	Learns feature representations
Dropout	0.3	Reduces overfitting
Dense layer 2	32 neurons, ReLU	Learns higher-level relationships
Dropout	0.2	Regularization
Dense layer 3	16 neurons, ReLU	Further feature transformation
Output layer	1 neuron, sigmoid	Produces fraud probability

The ReLU activation function is:

$$ReLU(x) = max(0, x)$$

The sigmoid output function is:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

The sigmoid output represents the model's estimated probability of a transaction belonging to the fraud class.

6.3 Training Parameters

Parameter	Value
Framework	TensorFlow/Keras
Model	Multilayer Perceptron
Optimizer	Adam
Loss function	Binary cross-entropy
Batch size	2,048
Initial maximum epochs	30
Hidden-layer activation	ReLU
Output activation	Sigmoid
Initial learning rate	0.001 by default
Class imbalance strategy	Balanced class weights
Early stopping	Validation PR-AUC
Learning-rate scheduler	ReduceLROnPlateau

The actual number of epochs completed is determined by the training history and early-stopping condition.

6.4 Loss Function

Binary cross-entropy is used to measure the difference between predicted probabilities and actual binary labels.

$$L = -\frac{1}{N}\sum_{i=1}^{N}[\, y_i \log(p_i) + (1 - y_i) \log(1 - p_i)\,]$$

Here, y_i represents the actual label and p_i represents the predicted probability.

Class weights modify the contribution of observations during training to account for class imbalance.

6.5 Hyperparameter Tuning

Basic hyperparameter tuning was performed by comparing the following learning rates:

- 0.001
- 0.0005
- 0.0001

The same network architecture and data split were used for comparison. Each trial was monitored using validation PR-AUC, with early stopping applied to limit unnecessary training.

The learning rate producing the best observed validation PR-AUC was identified as the best candidate among the tested settings.

Observed result: [Insert the tuning table and the best learning rate.]

The learning-rate comparison provides an initial assessment of training sensitivity. More extensive tuning could investigate batch size, network width, dropout rates, and other parameters.

6.6 Training Procedure

The model training procedure consists of:

1. Feeding standardized training features into the network.
2. Calculating predicted fraud probabilities.
3. Computing the weighted binary cross-entropy loss.
4. Updating model parameters using the Adam optimizer.
5. Monitoring training and validation metrics.
6. Applying early stopping when validation PR-AUC stops improving.
7. Reducing the learning rate when validation loss stops improving.

The trained model is then evaluated on the held-out test set.

7. RESULT VISUALIZATION AND INTERPRETATION

7.1 Evaluation Metrics

The model was evaluated using multiple classification metrics because fraud detection involves an imbalanced target variable.

Accuracy

Accuracy is the proportion of correctly classified observations.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

Precision measures the proportion of predicted fraud cases that are actually fraudulent.

$$Precision = \frac{TP}{TP + FP}$$

Recall

Recall measures the proportion of actual fraudulent transactions correctly detected.

$$Recall = \frac{TP}{TP + FN}$$

F1-score

The F1-score is the harmonic mean of precision and recall.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

ROC-AUC

ROC-AUC measures the model's ability to distinguish between the two classes across classification thresholds.

Average Precision

Average precision summarizes precision-recall performance across thresholds and is particularly informative for imbalanced classification.

7.2 Test-Set Evaluation Results

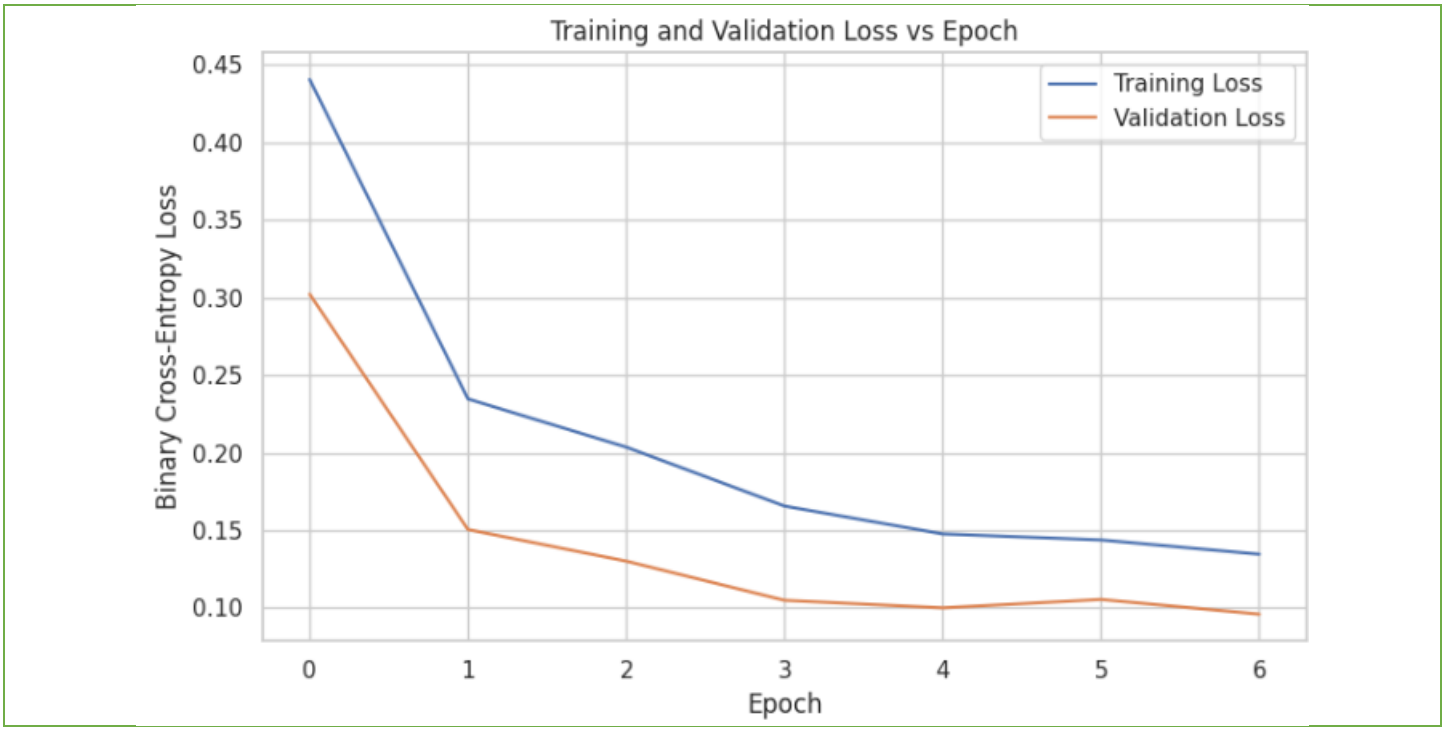
The model was evaluated using the held-out test set with an initial classification threshold of 0.5.

Metric	Observed value
Accuracy	[Insert actual value]
Precision	[Insert actual value]
Recall	[Insert actual value]
F1-score	[Insert actual value]
ROC-AUC	[Insert actual value]
Average precision	[Insert actual value]

These values must be copied from the evaluation output produced by the notebook.

7.3 Loss versus Epoch

Figure 8: Training and Validation Loss versus Epoch



The loss curve shows how the model's prediction error changes during training.

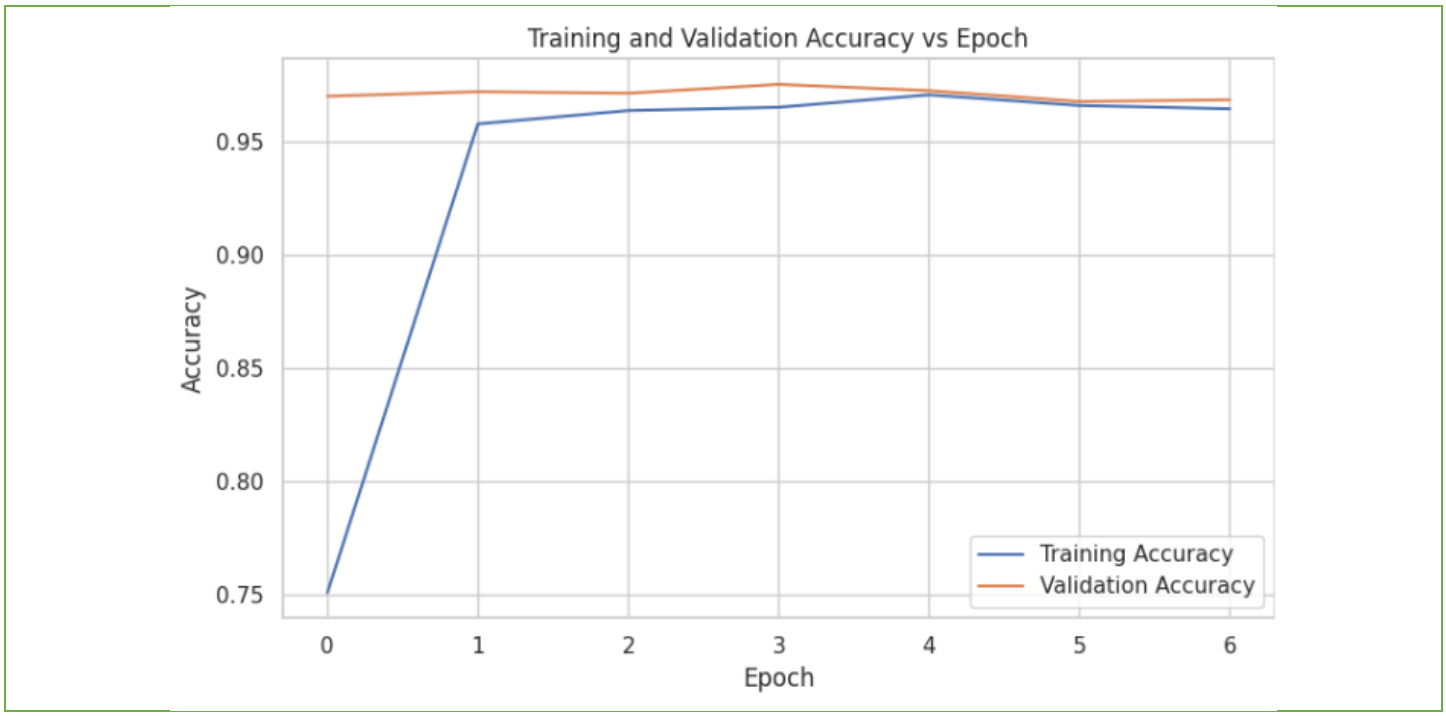
A decreasing training loss indicates that the model is fitting the training data. The validation loss helps determine whether this learning generalizes to unseen observations.

If training loss decreases while validation loss increases, this may indicate overfitting. If both losses remain high, the model may not have learned the patterns adequately.

Observed result: [Describe the actual training and validation loss trends.]

7.4 Accuracy versus Epoch

Figure 9: Training and Validation Accuracy versus Epoch



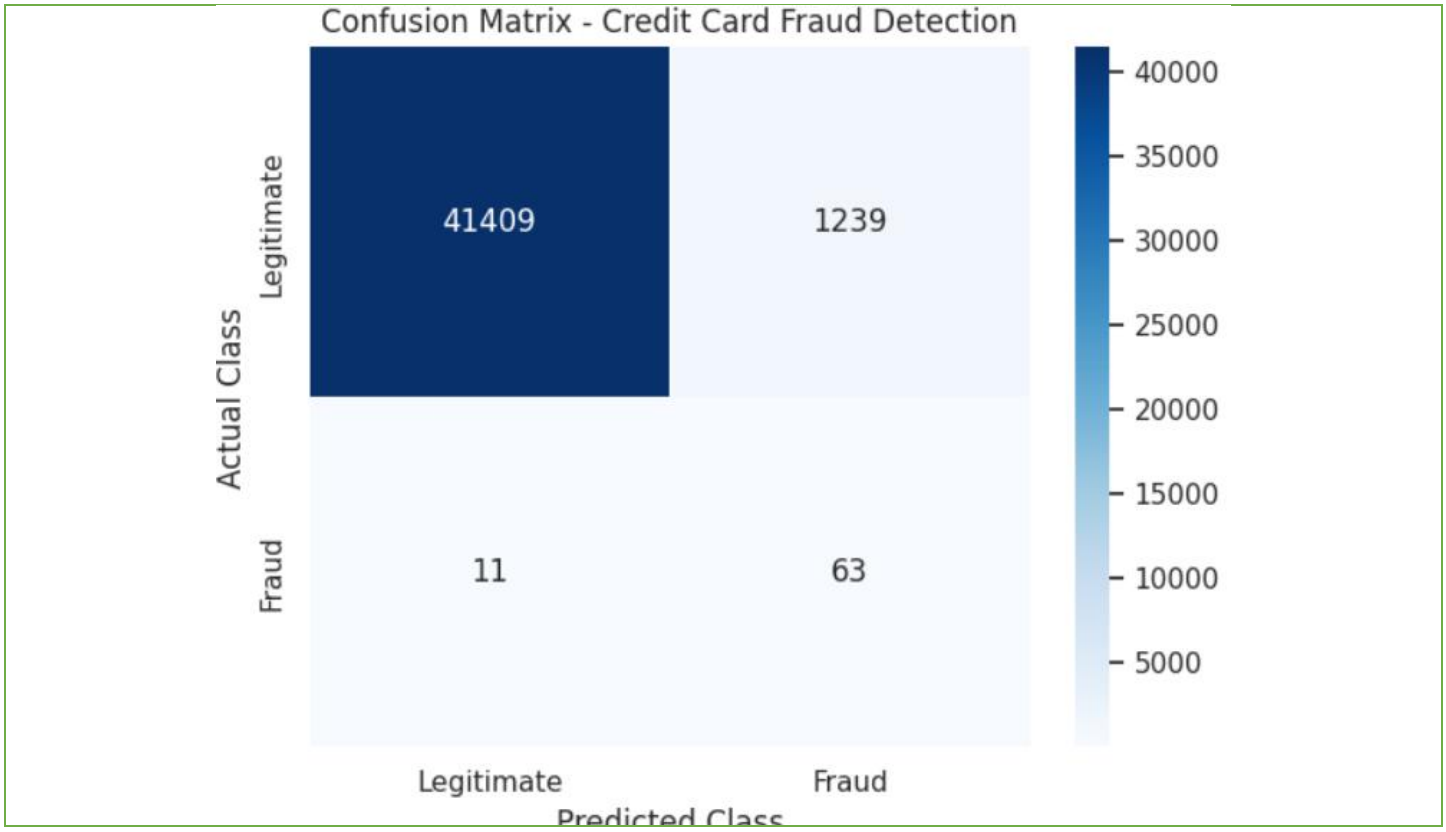
The accuracy curve illustrates the proportion of correctly classified observations during each training epoch.

Comparing training and validation accuracy helps assess learning stability and possible overfitting. However, accuracy must be interpreted alongside recall, precision, and PR-AUC because the majority class dominates the dataset.

Observed result: [Describe the actual accuracy trends.]

7.5 Confusion Matrix

Figure 10: Confusion Matrix of the MLP Classifier



The confusion matrix contains four outcomes:

- True negatives (TN): legitimate transactions correctly classified.
- False positives (FP): legitimate transactions incorrectly flagged as fraud.
- False negatives (FN): fraudulent transactions missed by the model.
- True positives (TP): fraudulent transactions correctly classified.

The matrix provides a direct view of fraud detection performance and the types of classification errors.

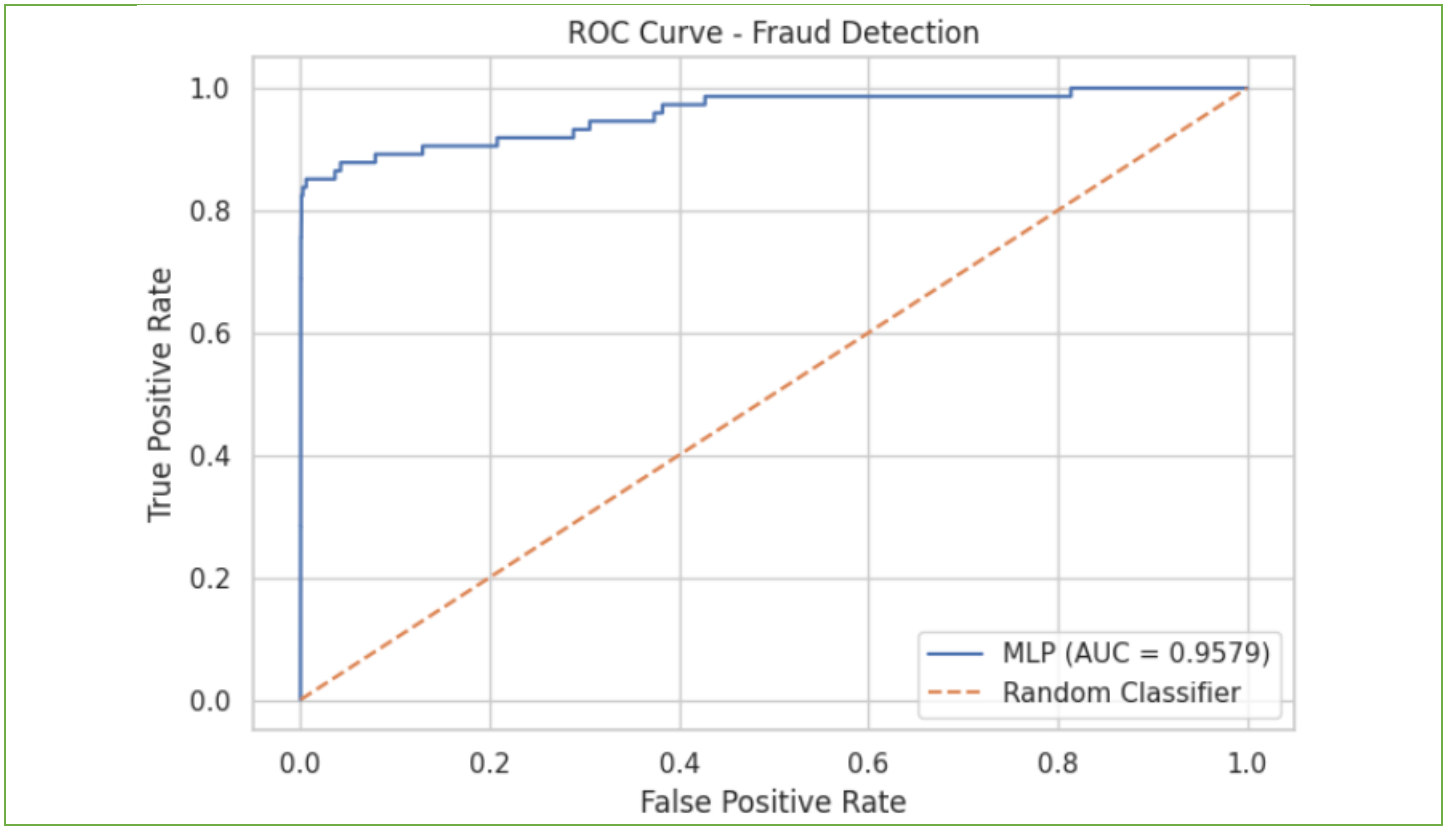
Observed result:

- True negatives: [Insert count]
- False positives: [Insert count]
- False negatives: [Insert count]
- True positives: [Insert count]

The relative number of false negatives and false positives helps evaluate the trade-off between missed fraud cases and unnecessary fraud alerts.

7.6 ROC Curve

Figure 11: ROC Curve of the MLP Classifier



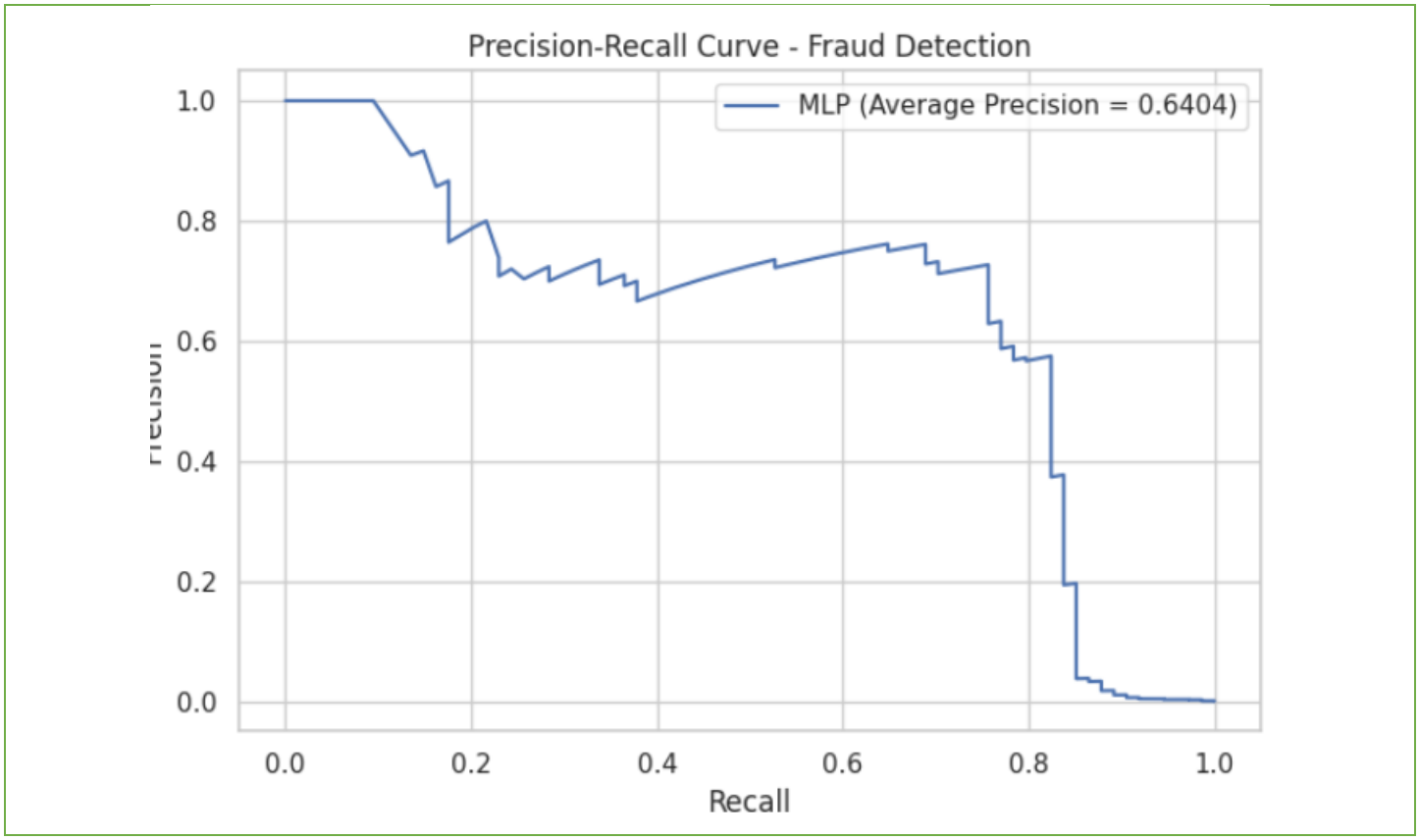
The ROC curve plots the true positive rate against the false positive rate at different classification thresholds.

A model with stronger class discrimination generally achieves a higher ROC-AUC. However, the practical usefulness of the model must also be evaluated through its precision-recall performance.

Observed result: [Insert the actual ROC-AUC and describe the curve.]

7.7 Precision-Recall Curve

Figure 12: Precision-Recall Curve of the MLP Classifier



The precision-recall curve shows how precision changes as recall varies across classification thresholds.

In fraud detection, increasing recall may identify more fraudulent transactions but can also increase false alerts. Average precision provides a summary of precision-recall performance.

Observed result: [Insert the actual average precision and describe the observed trade-off.]

7.8 Overall Discussion

The evaluation results should be interpreted together rather than relying on a single metric.

Accuracy provides an overall measure of correctness, while precision and recall focus on fraud identification and false alerts. The F1-score combines precision and recall, ROC-AUC measures discrimination across thresholds, and average precision summarizes precision-recall performance.

The training and validation curves provide evidence about learning behavior. The confusion matrix reveals the actual classification errors at the selected threshold.

The effectiveness of the MLP model is determined from the observed test metrics and visualizations. No claim of superiority over other models should be made unless a fair comparison has been performed.

8. CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

This project presents an end-to-end analysis of credit card transactions using Exploratory Data Analysis, statistical visualization, data preprocessing, and deep learning.

The dataset was examined to understand its structure, target distribution, transaction amounts, feature relationships, and potential outliers. Multiple visualizations were developed to study univariate distributions, class differences, and multivariate correlations.

The data was divided into training, validation, and test sets using stratified sampling. StandardScaler was fitted on training data and applied to the remaining subsets. Balanced class weights were used to address the imbalance between legitimate and fraudulent transactions during model training.

An MLP neural network was implemented using TensorFlow/Keras. Its training behavior was monitored using validation PR-AUC, early stopping, and learning-rate reduction. Basic learning-rate tuning was also performed.

The model was evaluated using accuracy, precision, recall, F1-score, ROC-AUC, average precision, and a confusion matrix. These measures provide complementary information about its ability to distinguish fraudulent transactions from legitimate ones.

The project demonstrates the importance of combining statistical exploration, meaningful visualization, appropriate preprocessing, and multiple evaluation metrics when developing a fraud detection model.

Final conclusion based on actual results: [Summarize the observed test metrics and state the main findings supported by your analysis.]

8.2 Challenges Faced

The main challenges associated with this project include:

1. Handling the substantial imbalance between legitimate and fraudulent transactions.
2. Interpreting anonymized features whose original business meanings are unavailable.
3. Identifying potential outliers without automatically treating them as incorrect data.
4. Preventing data leakage during data splitting and feature standardization.
5. Selecting evaluation metrics appropriate for rare-event classification.
6. Balancing fraud detection recall with the number of false alerts.

7. Selecting suitable hyperparameters and controlling overfitting.

8.3 Future Scope

The project can be extended in several ways.

1. **Threshold optimization:** Select a decision threshold using validation data according to the desired trade-off between fraud recall and false positives.
2. **Model comparison:** Compare the MLP with logistic regression, random forest, and other suitable classifiers using consistent data splits and metrics.
3. **Imbalance-handling methods:** Investigate resampling or other imbalance-aware methods using training data only.
4. **Explainable AI:** Apply appropriate explanation techniques to understand which features contribute to model predictions.
5. **Extended hyperparameter tuning:** Investigate network architecture, batch size, dropout, and learning-rate configurations.
6. **Real-time fraud detection:** Explore integration with streaming transaction systems and low-latency inference.
7. **Model monitoring:** Investigate data drift, changing fraud patterns, and periodic model evaluation.
8. **Deployment:** Develop an API or dashboard to demonstrate transaction risk scoring in a controlled environment.

These improvements could support further research into reliable, interpretable, and operationally useful fraud detection systems.

9. REFERENCES

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10. APPENDIX — SOURCE CODE

The project source code is provided in a Google Colab/Jupyter Notebook containing the following modules.

Appendix A: Dataset Loading

- Importing the required libraries.
- Loading the CSV dataset.
- Inspecting dataset dimensions and column names.
- Examining data types and statistical summaries.

Appendix B: Exploratory Data Analysis

- Missing-value and duplicate analysis.

- Class distribution and percentage calculations.
- Descriptive statistical analysis.
- Outlier analysis using the IQR method.
- Correlation analysis.
- Generation of EDA visualizations.

Appendix C: Data Preprocessing

- Separating input features and target labels.
- Stratified training, validation, and test splitting.
- Feature standardization.
- Calculation of balanced class weights.

Appendix D: Deep Learning Model

- Defining the MLP architecture.
- Compiling the model with Adam and binary cross-entropy.
- Training with class weights.
- Applying early stopping and learning-rate reduction.
- Comparing learning rates using validation PR-AUC.

Appendix E: Evaluation and Result Visualization

- Generating predicted fraud probabilities.
- Calculating classification metrics.
- Plotting training and validation loss.
- Plotting training and validation accuracy.
- Generating a confusion matrix.
- Plotting ROC and precision-recall curves.

Appendix F: Project Repository

The complete project will be maintained in a public GitHub repository containing:

- Project report in PDF or Word format.
- Source notebook (`.ipynb`).
- Dataset source link.
- README file containing the project overview, setup instructions, methodology, and results.